Exploring interpretations and consequences of statutory law through computational modeling: A case study in criminal justice reform

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1. Overview

The text and structure of statutory legislation is notoriously complex and specialized. This introduces confusion in interpretation even among experts and can leave ordinary citizens disempowered. Compounding the problem, legislation is dynamic in nature; it can be difficult for stakeholders to anticipate the consequences of proposed amendments. Our team has been exploring the application of computational modeling to a consequential, complex, and evolving piece of legislation: the State of Michigan's **Clean Slate** statute. (Michigan Legislature 2021 hereinafter "MCL 780.621") This law establishes conditions under which prior convictions can be removed from an individual's public record. Conviction **set-asides**, alternatively referred to as **expungements**, can have a tremendous positive effect in people's lives, but interviews conducted with legal experts who regularly deal with the statute have elicited difficulties they face with the statute's language: differences in interpretation, problems understanding the consequences, and a general sense of uncertainty that limits the full purpose of the statute.

We envision a "digital sandbox" for legislators, lawyers, judges, and citizens to define and explore the consequences of legislation. To this end, we have explored modeling the statute using the Alloy specification language (Jackson, 2012). An Alloy model provides both an formal representation of the statute's meaning and a means of performing automated analysis to search for unexpected "black swan" scenarios. In this paper, we motivate our work with excerpts from our expert interviews. We extract some of the concerns and edge cases brought up in their comments and illustrate them through scenarios generated by the Alloy analyzer.

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Related work. There have been a number of related efforts, leveraging computation to broaden access and engagement in regulatory and legal contexts: for example, automated forms that facilitate legal services such as eviction help Raby (2020), and auto-generated insurance contracts that are compliant with legal rules (CodeX, 2021). The Blawx system offers a block-based programming environment for generating legal services such as drafting a will or creating a lease (Morris, 2020). The creators of these projects argue that they lead to legislation that is clearer for all participants. Experiments with generating machine-readable regulations in New Zealand have indicated a speedup in interactions between government and business. Furthermore, machine-readable legislation allows stakeholders and public servants to test the impact of policy before they implement it (Darabi, 2018). Similarly, there are ongoing attempts to encode jurisprudence: for instance, court decisions that define and elucidate the regulatory environment (Ma et al., 2020). The preliminary success of these efforts has inspired this proposed project.

2. Clean Slate: History, implementation, practice

A criminal record, no matter how distant, creates barriers to employment, housing, and benefits. Consider the following experiences of recent expungement clients.

- Joe, 44 years old, takes whatever odd jobs he can find — fixing cars, repairing lawnmowers, shoveling snow — but he can't get a "real" job with benefits anywhere. This is because at age 18 — 24 years ago — he had a serious drinking problem, and while out drinking with friends, he stole a car, crashed it into a ditch then stole another car to get home. That's three felonies and an OWI (Operating While Intoxicated) just for that one night, in addition to other misdemeanors between 1994 and 2003. Although he has stayed away from drugs and alcohol for 20 years, the convictions on his record still keep potential employers away.
- Mandy, 50 years old, has been tutoring kids in her city and making sure they have meals on weekends, but she can't create an official non-profit to support this

work or use her social work degree to help anyone. This is due to a "retail fraud" charge for shoplifting nail polish when she was 19 and just entering college with no money.

- Sue is working on a nursing degree, but she is ineligible to work at a nursing home for the practical portion of her degree. 15 years ago, she bounced a check for about $30, then left town, missing a court date. Realizing her error two months later, she came back to see her charges increased to include both the misdemeanor fraud charge and a new felony absconding charge. With this criminal record, there has been no way for her to get regular employment.

- Bill, a truck driver for 30 years, was caught carrying a concealed weapon without a permit in his truck in 1997. He felt he needed it for protection at some of the rougher truck stops during the height of the crack cocaine epidemic. But he is now 64 and can't drive a truck any longer since his back went out. He is living on Social Security payments but is ineligible for subsidized housing or an apartment in a 55+ community because of the 1997 felony. After 30 years of hard work he is about to be homeless.

These barriers, which have been described as amounting to a "new civil death," magnify social and economic racial disparities (Alexander, 2010). A core part of this century's emergent criminal justice reform movement in the United States has been a search for effective ways to lessen barriers faced by people with criminal records. This effort is meeting with some success, with several U.S. states adopting one or more such policies in recent years.

Through its 2021 Clean Slate Act (MCL 780.621 et seq.), the State of Michigan is at the forefront of a movement to empower individuals by expunging or setting aside old criminal convictions. These records do not completely disappear, as the state police maintains a non-public record of the conviction that will be available if there is a new interaction with courts. However, the set aside records are not available to entities such as landlord's or employers during a background check. The Michigan State Court's Clean Slate website (2023) cites a University of Michigan study finding that those whose criminal records are set aside experience "a sharp upturn in their wage and employment trajectories" Prescott & Starr (2020). This view is certainly shared by the thousands of individuals who have applied for expungement relief since the new rules came into effect in April 2021.

In drafting the Clean Slate Act, the Michigan legislature set limits on which crimes and how many crimes can be expunged, and on how long an individual must wait after a conviction to become eligible. This was not a simple calculus, as legislators and citizens have conflicting and passionate opinions on all of these factors, and the resultant legislation is a complex document that reflects many compromises. The language has been subject to varied interpretations by attorneys and the judges tasked with ruling on these cases. Moreover, it is an evolving document as there have also been amendments to the law. For instance, the 2021 law initially excluded set-asides for operating a vehicle while intoxicated (OWI), but about a year later, February 2022, the statute was amended to allow one OWI conviction to be set aside. See MCL 780.621c(3) Other amendments are likely to come in the near future as legal aid and citizen groups work to lobby for differing protections.

3. Stakeholder experience and perspectives

Through the work of a dedicated group of legal aid agencies and citizens' groups, it seems that out of 2.8 million people with criminal records in Michigan about 250,000 individuals have been fully freed from the burden of a criminal record through the Clean Slate Act. Still, legal professionals who work on expungement cases testify to the confusion created by the statute's multiple interpretations. Our project team interviewed a number of these professionals, focusing on the following questions:

- To what degree is the statute, and its consequences, interpreted or understood in a unified way by judges and legal professionals across the state?

- To what degree do ordinary citizens interpret the statute correctly and understand its consequences?

We asked a small sample of attorneys to answer some Likert-scale questions and follow up with free comments on their experiences with the Clean Slate Act. Our respondents were the following:

- Respondent A is an attorney with a private practice in the Upper Peninsula of Northern Michigan. A has worked on over 100 set-aside applications through their private practice and through a contract with the state of Michigan's Michigan Works program.

- Respondent B is an attorney with a private practice in Northwest Michigan. B is involved with set-asides through the Networks Northwest Michigan Works program in which they have screened nearly 500 potential set-aside applicants and worked with approximately 150 qualifying applicants.

- Respondent C is an attorney with a private practice in Western Michigan, but whose primary work is in the legal services and nonprofit sectors. C has worked on more than 250 expungement cases and is currently the

Clean Slate Program Manager for Safe & Just Michigan, a non-profit organization dedicated to reducing "harm caused by both crime and unnecessary incarceration" (SJM, 2023).

- Respondent D is an attorney from South Central Michigan who works exclusively on expungement cases in the legal services sector. D has worked on over 150 set-aside applications and represented over 50 clients in multiple courts.
- Respondent E is an attorney from the Upper Peninsula of Northern Michigan and whose work is primarily in academia. E has worked with approximately 100 set-aside clients at various stages from application to court hearings through UP-MichiganWorks and has participated in various pro-bono screening events and fairs.

Our interviews started with a series of Likert-scale questions: How confident on a scale of 1-5 (1 being not at all confident and 5 being extremely confident) are you that:

1. *the statute is interpreted and understood in a unified way by legal professionals across the state?*
2. *legal professionals understand the consequences of the statute?*
3. *the statute is interpreted and understood in a unified way by judges across the state?*
4. *ordinary citizens interpret the statute correctly and understand its consequences?*

Likert scale results are presented below. Respondents generally felt confident that the consequences of the statute are understood by legal professionals in a unified way. However, they generally indicated low confidence in the unity of interpretation of the statute by both legal professionals and judges. They also overwhelmingly reported low confidence in the understanding of the statute by ordinary citizens.

Question	A	B	C	D	E
(1)	4	3	2	3	1
(2)	5	4	5	4	4
(3)	4	3	2	1	1
(4)	2	2	2	1	1

Consistently with the Likert scale results, the open-ended responses suggest that ordinary Michigan citizens have a limited understanding of the statute. This issue is particularly salient with those who seek expungements. Respondent B recalled their experience at set-aside clinics at fairs in which "[p]eople will come to us… and say 'I want to get my… [conviction] set aside so that if I get in trouble again, they'll treat it like a first offense'". Respondents indicated that those seeking expungements are not well served by official instructions, and that they rely primarily on unreliable sources for information. As Respondent D indicated, reading complicated legal documents is a barrier that causes many to give up or not even begin the process:

> "[The statute's] language is very confusing… It's huge. I have to read it nearly every day… [and] I'm constantly learning new things. I find that it's very, very difficult… for someone who has a felony conviction to figure out even what they need to do, and the instructions attached to the application are not that clear even if you were to follow them exactly."

4. The statute and judicial interpretation

The Clean Slate Act, like most legislation, has plenty of ambiguity and is therefore subject to the interpretation of lawmakers, judges, and legal professionals. As a primary example of this, it is unclear how statutory time constraints should apply. For the purposes of this paper we will consider two major approaches to interpretation of the statutory language as identified by interviewees involved in representing clients in the expungement process.

Statutory basics: Time periods. The statute mandates that an application can be filed only after all prison and parole conditions have been completed plus a seven year wait for records with more than one felony, five year wait for records with just one felony, or a serious misdemeanor or Operating While Intoxicated (OWI) charges, and finally just a three year wait to file for most other misdemeanor convictions. This section then goes on to state that for a judge to grant the set aside it must be shown that the "the applicant has not been convicted of any criminal offense during the applicable time period required." (MCL 780.621d (4)(c), emphasis added). But the statute does not specify exactly what the applicable time period is. Should it be the date of the offense looking forward three, five, or seven years, or should it be the current date looking back three, five, or seven years — or perhaps both need to be met?

More than three felonies and "One Bad Night". The statute can also be read to hold that a person with more than three felonies can be precluded from having any of their felony convictions set aside: "a person convicted of one or more offenses but not more than a total of three felony offenses... may apply to have all of his or her convictions… set aside". MCL 780.621(1)(a). This could mean that an upper limit of three convictions can be set aside, or it could

mean that an individual with 4 convictions will not get any of them set aside.

In any event, the application process has a special process/exception for the all-too-common situation where several offenses occur as part of the same transaction within a 24 hour period. Consider a case where in 2005 an applicant and a friend robbed a liquor store. They were convicted of five felonies: 1) breaking and entering, 2) theft of over $200 worth of merchandise, and when the police arrived, 3) fleeing and eluding law enforcement, 4) conspiring with others to do so, and 5) Operating While Intoxicated as they tried to get away. The fact that the applicant had 4 felonies in one day would ordinarily make them, and all of their convictions NOT eligible for expungement. MCL 780.621(1)a. But the statute's "One Bad Night" counting rule means that the court could count all these felonies as "a single felony" for the purpose of the statute. See MCL 780.621b(1). Thus, if there were no other incidents, these acts would all be expunged under the application process. Indeed, the applicant could have two more felonies and still have all of the offenses set aside.

Common judicial interpretations. Our interviews indicate a range of interpretations by legal professionals and judges across the state in several key parts of the statute. In particular, our interviewees described ambiguity in the statute's language on waiting periods for expungements:

> "Say you have a felony or two felonies which will require a seven-year waiting period, there is one school of thought that says, hey, if you are trying to get that felony set aside and seven years have passed since the later of those two felonies, then you are good to go on that application… The other school of thought says, if you have committed crimes in the meantime, you should wait until the waiting period for that particular crime [is up]."

Another interviewee experienced an even more conservative interpretation of waiting periods:

> "The prosecutor argued that since the former convict had a felony in 2000 and another one two years later [in 2002]... he didn't wait seven years [between felonies]. And so, the prosecutor's interpretation was that you had to wait… five to seven years between each felony to get any of them expunged."

Interviewees also described issues with interpreting the special rules for expunging OWI (operating while intoxicated) offenses:

> "That section, [which] became law in March of 2022… doesn't say that an individual is not eligible [for getting a single OWI expunged] if they've had more than one offense. But the office of the Attorney General has been opposing situations in cases where somebody has had a first time OWI, followed by a subsequent OWI."

The following scenario demonstrates how differing interpretations of this "applicable time period" rule might impact an application. For the purpose of simplicity we are focusing here on just two dominant methods of interpretation seen by the practitioners interviewed. For example, consider reviewing the record of an applicant in 2025 with the following convictions:

1. Felony (2006)
2. Misdemeanor OWI (2008)
3. Felony (2010)
4. Misdemeanor (2022)

The court's decision as to what offenses to set aside depends greatly on how the particular judge reads the statute. Interpretative approach 1 (counting forward from the date of conviction) is what attorneys view as the narrowest interpretation of the applicable time period provision. With this approach, the Court looks forward from the date of the conviction release to see if there is a subsequent conviction within three years (for a misdemeanor conviction), five years (for a single felony, OWI, or serious misdemeanor) or seven years (for two or more felonies) to determine if the prior conviction can be expunged.

In our sample scenario, counting **forward** would allow the 2022 conviction to be expunged in 2025. The 2010 conviction could be expunged in 2025 as well, but when considering the 2008 and 2006 convictions, the judge would conclude they can never be expunged because of the fact that there are intervening convictions within the "applicable time periods."

It is generally thought that most offenders commit their offenses within the span of a few years (often before their mid-20's), and the counting forward approach presents obstacles to expungement in such cases. An individual who commits three felonies eight years apart (over a 24 year period) may have them all expunged, but if the same three felonies occurred between ages 19 and 24 (within a five year time span) twenty years ago, only the most recent felony would be expungeable.

For Interpretive Approach 2 (counting **backward** from the date of application), the court views the start of the applicable time period as the date of the application for set aside

and counts backwards. This method is used by many courts and arguably justified by the Court's need to find the applicant is currently of good character and an asset to the community. In this approach the applicant will not get any convictions set aside until 2025, three years after the 2022 misdemeanor — but in 2025 all the convictions could be set aside. This interpretation uses the current date (date of application) to start the look back period, but if there are any recent convictions in the past three years the court would not go back any further and the applicant would need to wait until the most recent event can be cleared.

5. Automation to the rescue?

In crafting the Clean Slate law, the Michigan Legislature showed foresight recognizing that, with over two million citizens with criminal records, the individual application process would be extremely time consuming for the courts, and both difficult and expensive for the average citizen to pursue. In response, the Legislature mandated creation of an automated process to relieve this pressure on courts. (MCL 780.621g). The Michigan State Police (MSP), required by the law to coordinate creation of a process for automatic expungement (MSP, 2023; Sahouri, 2023a). The automated system, which the MSP refers to as a "Rules Engine", went into operation in April 2023.

The automated system developed by the MSP is undeniably a step in the right direction, as it is so much more efficient and removes the burden on the courts and citizens. In the first year the statute was in full effect and before any automated process was available, estimates are that out of about 2.8 million people with criminal records, the MSP processed about 16,283 applications to set aside convictions (MSP, 2023). Since automation was introduced in April 2023, iit is estimated that 252,000 citizens have had their records cleared and 1.2 million offenses have been expunged. Stebbins (2023b) Thus, even though a wider variety of offenses are eligible for set aside sooner under the application process, that process remains difficult, time consuming, costly and confusing to most citizens, and the automation process is currently doing much of the work for Michigan's Clean Slate law.

Thus it would seem from the basics outlined in the statute that in a given case, what could be set aside automatically would be a simple subset of what a citizen would be eligible to see set aside by individual application. However, in effect and practice, it is not a simple subset, and indeed the automatic process created by the MSP has allowed for both more and fewer expungements than courts have allowed for, which further illustrates some of the varying interpretations of the statute by Judges, the Michigan State Police as an Agency, and the Attorney General's office.

Without more public education, resources, and transparency by the MSP as to how the rules engine works, most individuals who do benefit by getting some offenses set aside via the automatic process will never know that they could have other offenses set aside by application to the courts. Indeed, they may never know that they had any offenses set aside automatically, as they have to initiate that query and pay to generate an electronic copy of their record through IChat (Internet Criminal History Access Tool). Of even greater concern is the risk that many, including attorneys and judges, will believe that since the automated process for set-asides have occurred, any remaining offenses are somehow more serious and either un-expungeable or should be more carefully scrutinized before additional asetasdes are applied for or ordered.

Inside the Rules Engine. In creating the rules engine for the automatic process, the MSP had somewhat different constraints (e.g., seven and ten year wait periods) and a longer list of offenses that were not eligible under the automatic process. This left the MSP an opportunity to encode its own interpretation of the law that may have been based on either ideological or pragmatic considerations rather than concerns about statutory interpretation or fealty to an interpretive methodology. Practitioners have apparently had little involvement or information as to how the MSP developed its rules engine, and there is no public statement. The information available currently comes from a webinar with legal professionals and a PowerPoint document developed by the MSP.

The presentation describes the operation of the Rules Engine as an iterative process. It first looks at the oldest offense (Offense 1), then looks forward for any offense within x years (seven for misdemeanors and ten for felonies). If there is no offense for x years back, the engine will set aside Offense 1. If on the other hand there is an offense (Offense 2) within the x-year period, Offense 1 is not set aside for the time being, and the process continues with Offense 2. If Offense 2 is set aside, the process will eventually return to Offense 1 and (in the absence of any other remaining conviction in the x-year period) will set it aside.

The working process of the Rules Engine was explained in the Powerpoint Presentation offered by the MSP as follows:

> The rules engine will start by looking at the oldest offenses in the CHR and apply the criteria provided in MCL 780.621g, e.g. Individual A's criminal history:
>
> 1. 1995 – misdemeanor
> 2. 2003 – misdemeanor
> 3. 2011 – one felony
> 4. 2022 – one misdemeanor

> In this scenario, convictions 1 through 3 will be set aside by the rules engine on first pass through the system. Conviction 4, on July 8, 2022, will not be automatically set aside because seven years have not passed yet. It will be eligible to be automatically set aside on July 9, 2029, provided the set aside is permissible at that time per MCL 780.621g.

In the next example, the MSP examined the following criminal history:

> Consider Individual B with the following history:
>
> 1. 2000: misdemeanor
> 2. 2009: misdemeanor
> 3. 2010: felony
> 4. 2010: misdemeanor
> 5. 2011: misdemeanor
> 6. 2012(a): misdemeanor
> 7. 2012(b): felony (same transaction) (felony entered last)

The presentation explained,

> The order of setting aside via the rules engine would be: (1) 2000 misdemeanor. Because there are intervening offenses per MCL 780.621g (6) (c), the rules engine cannot set aside 2-7 in order. In that scenario, the rules engine will then apply MCL 780.622: "upon the automatic setting aside of a conviction under section 1g, the applicant . . . is considered not to have been previously convicted. . . " So the order would be to next set aside (7), 2012b felony, (6) 2012 misdemeanor, (5) 2011 misdemeanor, (4) 2010 misdemeanor, (3) 2010 felony 2 In the end, only offense (2) the 2009 misdemeanor would remain on the record due to the maximum allowance being reached" (MSP, 2023).

However, the presenters did not discuss explicitly how the Rules Engine would process a similar record where there are intervening offenses that cannot be set aside (for example an OWI offense). But, according to the explanation of the process as offered, if offense 5 was an OWI and could not be expunged automatically, then since offense 5 would remain on the record, neither offenses 2, 3 or 4 would be eligible for expungement either. Thus, the current operation of the Rules Engine implies that the large list of crimes that cannot be expunged automatically will have a significant impact on the availability of the benefits of the clean slate program. This has been borne out by the current results, where the estimated 400,000 citizens who could be conviction-free after April 11, 2023 MSP (2023) has been reduced to 252,000 (Sahouri, 2023b).

Table 1 below sets out some sample scenarios for individuals and how their records would be impacted by the MSP Rules engines and other Interpretive methods practitioners have experienced in the Michigan courts.

Table 1:

Scenario Misd=Misdemeanor Fel=Felony SA=~~SetAside~~ NSA= NotSetAside Italics=(SA-eligibleYear) Date	Automatic MSP "Rules Engine" Process as described.	Interpretation 1 (Look forward from date of conviction)	Interpretation 2 (look back from application)
Ind A			(all SA in 2025)
1. 2022 Misd	2022 Misd (SA2029)	2022 Misd (SA 2025)	2022 Misd
2. 2011 Felony	2011 ~~Felony-SA~~	~~2011 Felony (SA)~~	2011 Felony
3. 2010 Misd	2010 ~~Mist ~ SA~~	2010 Misd (NSA) aside)	2010 Misd
4. 2003 Misd	2003 ~~Misd-SA~~	~~2003 Misd (SA)~~	2003 Misd
5. 1995 Misd	1995 ~~Misd-SA~~	~~1995 Misd (SA)~~	1995 Misd
Ind B			
1. 2000 Misd	~~2000 Misd~~	~~2000 Misd~~	~~2000 Misd~~
2. 2009 Misd	2009 Misd (NSA)	2009 Misd- NSA	~~2009 Misd~~
3. 2010 Felony	~~2010 felony~~	2010 Felony-NSA	~~2010 Felony~~
4. 2010 Misd	~~2010 Misd~~	2010 Misd-NSA	~~2010 Misd~~
5. 2011 Misd	~~2011 Misd~~	~~2011 Misd-NSA~~	~~2011 Misd~~
6. 2012 Misd	~~2012 Misd~~	~~2012 Misd~~	~~2012 Misd~~
7. 2012 Felony	~~2012 Felony~~	~~2012 Felony~~	~~2012 Felony~~
Ind 2 with OWI (oper under influence)			
1. 2000 Misd	~~2000 Misd~~	~~2000 Misd~~	~~2000 Misd~~
2. 2009 Misd	2009 Misd	2009 Misd	~~2009 Misd~~
3. 2010 Fel-May	2010 Felony	2010 Felony	~~2010 Felony~~
4. 2010 Misd-Oct	2010 Misd	2010 Misd	~~2010 MisdOWI~~
5. 2011 Misd OWI	2011 Misd OWI	2011 Misd OWI	~~2011 Misd~~
6. 2012 Misd	~~2012 Misd~~	~~2012 Misd~~	~~2012 Misd~~
7. 2012 Felony	~~2012 Felony~~	~~2012 Felony~~	~~2012 Felony~~

As can be seen in Table 1, the variation in results is significant. Thus, while it is not clear if MSP's Rules Engine constraints are borne of pragmatic concerns with regard to interpretation of the statute's requirement or the programming design, as it seems that in practice the MSP Rules Engine's circular process does not allow for expungement of eligible offenses that are in close proximity to what are ineligible offenses (such as OWI) under the automatic set aside provision. As a result, while the automatic process has been extremely beneficial to many, there are still many citizens with records that will need to go through the courts if they wish to truly have a "clean slate".

6. Modeling and analysis with Alloy

To explore the potential of computational modeling for exploration and learning in the context of Clean Slate, we use the modeling language Alloy and the Alloy Analyzer to provide hands-on experience with automated feedback (Jackson, 2012). An Alloy model captures real-world concepts and entities as abstractions, called signatures, and enables designers to define relations between signatures. Designers can then specify a set of mathematical constraints on signatures and relations. Constraints may be expressed in Alloy as first-order formulas, as relational expressions,

or a combination of both. In addition, the temporal operators of Linear Temporal Logic (LTL) are built into the language. The Alloy Analyzer searches within designer-specified bounds for instances satisfying an Alloy model, through a SAT-based constraint solver. If a satisfying instance is found, the analyzer displays it graphically; otherwise, it identifies a minimal unsatisfiable core of the model. Alloy has been used in a wide array of application areas, including security, computer architecture, program verification, and model driven development (Jackson, 2019).

In our model, instances consist of temporal sequences of events (convictions and expungements), each with an associated date. **Event** and **Date** are defined as signatures in our model. Convictions are subclassed into misdemeanors and felonies, with further subclasses that are relevant to the language of the statute.

```
-- An event is a conviction or an expungement
abstract sig Event {date: one Date}
-- each event has an associated date
-- A conviction is a felony or a misdemeanor
abstract sig Conviction extends Event { }
abstract sig Felony, Misdemeanor extends Conviction { }
-- Special types of felony: assaultive, ten-year
sig AssaultiveFelony, TenYearFelony in Felony { }
-- Special type of misdemeanor: OWI (Operating While Intoxicated)
sig OWI in Misdemeanor { }
-- Each expungement has a set of related convictions
-- note: multiple convictions may be expunged in a single event
sig Expungement extends Event {con: some Conviction}
```

At each point in time, a set of events may occur, represented by the variable signature **now**. We allow simultaneous convictions to capture the possibility of multiple criminal activities happening in "one bad night". Using Alloy's built-in temporal operators **eventually** and **after**, we define a happens-before relation over the **Event** signature. The **eventually** operator means that some condition holds now or at some future point of time, and **after** means that some condition holds in the next point of time. The predicate **hb** behaves as a Boolean function that returns true if the constraint inside it holds, and returns false otherwise. Note that **hb** is irreflexive; i.e. it holds for pairs of events **[e1, e2]** where **e1** strictly precedes **e2**.

```
-- now indicates the current event
var sig now in Event { }
-- The strict happens-before relation for events
-- (i.e. no reflexive pairs)
pred hb[e1, e2: Event] {
    eventually (e1 in now and after eventually e2 in now)
}
```

In order to capture the waiting-period aspects of the statute, we introduce a linearly ordered set of dates, with binary relations to capture the relevant waiting periods mentioned in the statute. For example, the binary relation **withinThree** captures all ordered pairs **d1->d2** where **d2** is within a three-year span of **d1**.

```
open util/ordering[Date]    -- Dates are linearly ordered
-- Associated with each date is the events occurring
-- within 3/5/7 years of this date
sig Date { withinThree, withinFive, withinSeven: set Date }
```

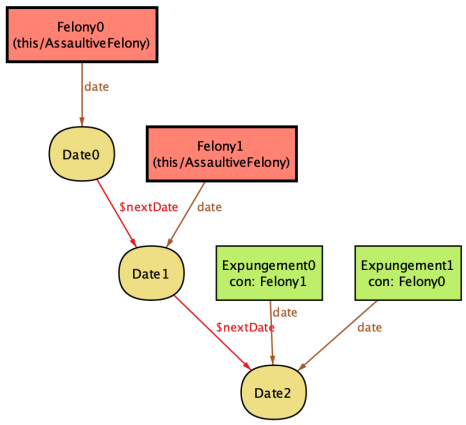
We express the statute's restrictions on expungements as temporal logic formulas. For instance, Section 1,1b states that no more than two assaultive felonies may be expunged. We define a predicate to capture the occurrence of three (or more) assaultive felonies (using our previously defined happens-before relation **hb**) and then assert that in such cases the third assaultive felony is not expunged.

```
-- Does the assaultive felony af occur after two preceding
-- assaultive felonies?
pred afterSecondAssault[af: AssaultiveFelony] {
    some af1, af2: AssaultiveFelony | af1 != af2
        and hb[af1, af] and hb[af2, af]
}
-- No more than two assaultive felonies may
-- be expunged (Sec. 1, 1b).
pred sec1_1b {
    no af: AssaultiveFelony | afterSecondAssault[af]
        and expunged[af]
}
```

We can verify the expungeability of a given conviction by setting up a scenario that instantiates the relevant events and their relative time orderings. For instance, consider a simple scenario where expungements are sought for a misdemeanor conviction followed by a felony conviction. We define a predicate capturing the details of the scenario and instructing the Alloy Analyzer to find a satisfying instance:

```
pred show {
    some af1, af2: AssaultiveFelony |
        hb[af1, af2] -- af1 happens first
            and af1.expunged and af2.expunged -- both expunged
}
run show for 4 Event, 3 Date
```

Note that we can specify upper bounds on the sizes of the signatures **Event** and **Date**. In this case, there is a satisfying instance (in fact, many satisfying instances) with only four distinct events and three distinct dates. The Analyzer gives a visualization of the instance as a sequence of states, with the initial state displayed below:



Although the instance involves three states, the initial state actually provides all the information needed to understand the scenario, due to the timestamping provided by the

date relation. Here, **Felony0** occurs first, followed by **Felony1** (more than seven years later), followed (more than seven years later) by an expungement of both **Felony0** and **Felony2**.

If we attempt to extend the scenario with a third expunged assaultive felony, Alloy responds with a failure message and an indication that the scenario is incompatible with the predicate **sec1_1b**:

```
pred show {
    some af1, af2, af3: AssaultiveFelony |
        hb[af1, af2] and hb[af2, af3]
        and af1.expunged and af2.expunged and af3.expunged
}
run show for 6 Event, 6 Date

Executing "Run show for 6 Event, 6 Date"
No instance found. Predicate may be inconsistent.
```

In addition to simple expungement testing for specific scenarios, the Alloy analyzer allows us to explore other questions regarding interpretations of the Clean Slate Act. For instance, we can express the difference between the "backward" and "forward" interpretations of waiting periods, described earlier, in terms of different predicates.

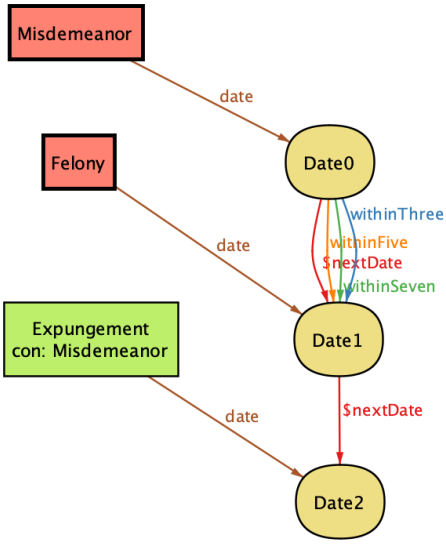
```
pred backwardWaiting {
    no x: Expungement |
            (some m: Misdemeanor | x.date in m.date.withinThree)
            or (some f: Felony | x.date in f.date.withinFive)
            or (some f: Felony | interveningFelony[f, x]
        and x.date in f.date.withinSeven)
}

pred forwardWaiting {
    all m: Misdemeanor |
        m.expunged implies no c: Conviction - m |
            c.date in m.date.withinThree
    all f: Felony | f.expunged implies no c: Conviction - f |
        c.date in f.date.withinFive
    all f1: Felony | f1.expunged implies no f2: Felony - f1 |
        f2.date in f1.date.withinSeven
}
```

The **backwardWaiting** predicate asserts that for any expungement, there is no preceding conviction in the "waiting period" (three years for a misdemeanor, five years for a single felony, and seven years for a felony with additional intervening felony). In contrast, the **forwardWaiting** predicate asserts that for any conviction, there is no following conviction in the waiting period.

We can view the differences between these two interpretations by instructing the Alloy Analyzer to find instances that satisfy one predicate but not the other.

```
pred show {
    backwardWaiting
    not forwardWaiting
}
run show for 3 Event, 3 Date
```

This will search for instances satisfying the backward interpretation of waiting periods but not the forward waiting interpretation. Note that upper bounds for the sizes of the **Event** and **Date** signatures are given; here we can find an instance with only three distinct events and three distinct dates. The Analyzer indicates the existence of a satisfying instance and provides a visualization.



The **Expungement** at **Date2** expunges the **Misdemeanor** at **Date0**. Note that **Date1** is within three years of **Date0**, but **Date2** is more than seven years after **Date1**.

This is indeed compatible with the backward interpretation of waiting periods, since **Expungement** takes place more than seven years after the previous **Felony** conviction. (In fact, under the backward interpretation, only three years since **Felony** would be sufficient.) On the other hand, it is not compatible with the forward interpretation, since Misdemeanor is followed by another conviction, **Felony**, before the waiting period of three years has expired.

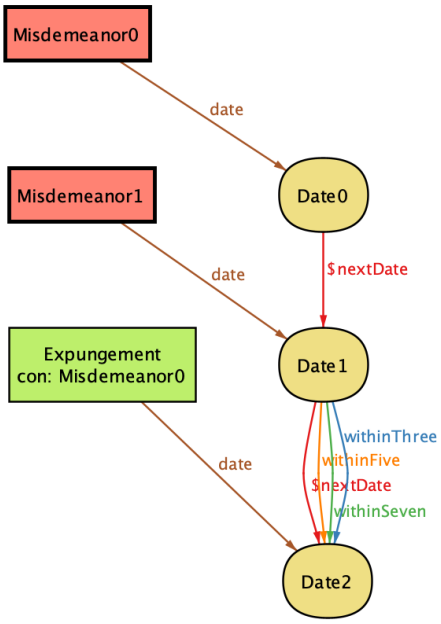
Alternatively, we can search for instances satisfying the forward waiting interpretation but not the backward waiting interpretation.

```
pred show {
    not backwardWaiting
    forwardWaiting
}
run show for 3 Event, 3 Date
```

The Analyzer indicates the existence of a satisfying instance and provides a visualization.

In this instance, **Misdemeanor1** takes place more than seven years after **Misdemeanor0**, so it does not constitute an obstacle to expungement of **Misdemeanor0** under the forward-looking interpretation. But **Expungement** does occur within three years of **Misdemeanor1**, which runs contrary to the backward-looking interpretation.

We can also use the Alloy Analyzer to clear up misconceptions regarding the consequences of the statute. For instance, it may be surprising to discover that an individual



may have more than three felonies expunged. At first reading, this appears to contradict Section 1,1a of the statute, which states, "No conviction may be expunged after three or more felonies". The relevant code from our Alloy model is given below.

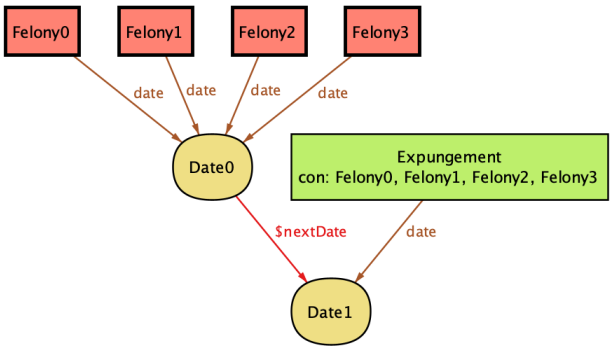
```
-- Does conviction c occur after three preceding felonies?
pred afterThirdFelony[c: Conviction] {
    some f1, f2, f3: Felony |
        hb[f1, f2] and hb[f2, f3] and hb[f3, c]
}
-- No conviction may be expunged after three
-- or more felonies (Sec. 1, 1a).
pred sec1_1a {
    no c: Conviction | afterThirdFelony[c] and expunged[c]
}
```

The key insight in our model is that multiple events can happen simultaneously, but since the **hb** relation is irreflexive, the **afterThirdFelony** is only counting sequences of felonies that happen at different times. This is compatible with the One Bad Night rule, which intentionally allows for multiple offenses that occurred at the same time to be counted as one offense for Section 1,1a and other clauses that "count" offenses.

We can instruct the Alloy Analyzer to search for instances with four expunged felonies.

```
some f1: Felony | some f2: Felony-f1 |
    some f3: Felony-f1-f2 | some f4: Felony-f1-f2-f3 |
            f1.expunged and f2.expunged
            and f3.expunged and f4.expunged
```

The Analyzer finds an instance involving a single "bad night" with four felonies, all of which are expunged at the same time.



7. Conclusion: Towards a digital expungement sandbox

Our investigation of Michigan's Clean Slate statute has revealed significant barriers for ordinary citizens, confusion even among legal experts, and unresolved differences in interpretation between lawyers and judges. Our goal is to increase participation, collaboration, and transparency among the stakeholders in Clean Slate, and ultimately other examples of statutory law. The research objective of this project is to develop a transformative, technology-empowered approach to inter-stakeholder communication, guiding stakeholders by formalizing statutory language, defining terms carefully and unambiguously, and anticipating the consequences of the proposed rules — including "black swan" scenarios (Kolt, 2023).

The Rules Engine of the Michigan State Police provides a useful counterpoint. The Rules Engine is essentially a black-box interface to the Clean Slate statute, indicating which convictions may be expunged within a given scenario, according to a fixed interpretation (which happens to differ in important ways from the interpretations used by judges), but providing relatively little in terms of explanation. In contrast, we envision a digital sandbox for experimenting with statutory law — a tool that not only allows users to explore consequences of existing law under a set interpretation, but also compare the consequences of different interpretations, and even create and explore the consequences of extensions or amendments to existing law.

A key component of such a sandbox will be a toolbox of predefined elements that users can use to build their own statutory regulations. To that end, it is important to identify common patterns within statutory law that we can capture with reusable code modules. The timing constraints in the statute can be classified into several categories of patterns, which are quantified in terms of the amount of time elapsed between events. Specifically, one can list at least the following types of patterns:

- absence of events in some future time interval (e.g., no expungement occurs within at least x years from completing the sentence of a crime);
- absence of events in some past time interval (e.g., if two crimes are committed in different occasions, then the second one can be expunged if at least x years have past since the first one);
- length of time interval between events (e.g., once a misdemeanor is served, it cannot be expunged until three years), and
- existence of an event between two other events (e.g., Is there an intervening felony between an expunged felony and 7 years before it?)

Dwyer (1999) provide specification patterns that capture the qualitative versions of the aforementioned constraints in that there is no specific amount of time between the occurrence of events. For instance, the "Existence Pattern" Dwyer (1999) specifies whether some event P exists between two other events Q and R, but does not stipulate any constraint on how far Q and R should be from each other. As another example, the "Precedence Pattern" requires that some event S occurs after another event T (e.g., the expungement of a crime always occurs after its term is served). To this end, we plan to investigate the use of real-time specification patterns Konrad and Cheng (372 – 381) in formal and visual specifications of timing constraints.

Another important question is the choice of the specification language. In this study, we have used Alloy due to its modeling power and automated scope-based analysis. While the Alloy language does provide a relatively elegant and compact model of the statute, the expressive power and flexibility of the language can potentially be a burden for non-experts trying to understand the code. Also, dealing with numeric time bounds is not particularly easy in Alloy. However, we believe that no single formal language will be a silver bullet for the specification and analysis of the statutory language. To address this problem, we will study the use of other formal languages (e.g., Metric Temporal Logic - MTL) in combination with Alloy. For example, MTL provides time-constrained versions of the temporal operators such as eventually, always and after. Such time-constrained operators will be very useful in exact specification of timing restrictions of the statue. Moreover, we plan to use visual specification languages to simplify the understandability of formal specifications for layman stakeholders. For instance, we will study how the Real-Time Graphical Interval Logic Moser et al. (1997) can help us visually represent the timing constraints and other queries on formal models.

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